

Class 3: XG Architecture Basics

RAN, Core, and O-RAN

Xin Liu

Florida State University
xliu15@fsu.edu

CIS 4930/5930 Future Edge Networks
xinliulab.github.io/FSU-CIS4930-CIS5930-Future-Edge-Networks

What does the icon actually tell you?



Ericsson active 5G antennas

Turn off Wi-Fi for ten seconds.

What appears: LTE, 5G, 5G+, or 5G UW?

- **LTE:** 4G radio access.
- **5G:** 5G radio is available.
- **5G+:** AT&T mid- or high-band.
- **5G UW:** Verizon C-band or mmWave.

LTE = Long Term Evolution mmWave = millimeter wave

Q: Does that icon describe the radio, the core, or the whole network?

A: Only the **radio** — the link between your phone and the tower — and even that is filtered through carrier branding. It says nothing about the **core**, the machinery behind the towers that knows who you are and decides where your traffic goes. And nothing about the speed you get.

Somebody paid about \$50 for that icon this month

What a line costs

- Entry unlimited: \$45–\$51
- Same price at 2 GB or 60 GB
- Hotspot only on some plans

Entry unlimited, one line, autopay required, before taxes and fees. August 2026.

What is not on the bill

- A price per gigabyte
- A speed you are owed
- Which radio you get

Q: If your bill does not change when your usage doubles, what are you buying?

A: Access, not data. You are buying the right to attach at all, which is why carriers compete on bundles instead of on megabits.

The bills are flat. The build is not.

What one line brings in

About \$56.57 a month per postpaid phone line.

Down 0.3% in a year — and roughly flat for three.

Both columns are AT&T, from its Q4 2025 and Q2 2026 SEC filings. Revenue per line is one carrier's; the pattern is the industry's.

ARPU = Average Revenue Per User capex = money spent building things

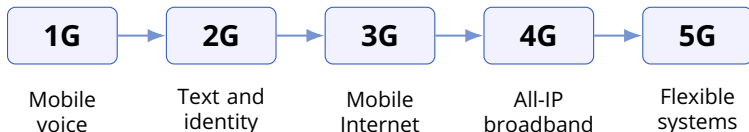
What the build costs

\$23–24 billion a year, guided through 2028.

Last quarter: \$5.7 billion, against \$4.9 billion a year earlier.

Flat revenue per line, rising spend. How long does that work?

Each generation made a new habit feel normal



The app was visible.

The architecture underneath was not.

The first mobile call was basically a flex



Martin Cooper with a 1973 DynaTAC prototype

Martin Cooper called his rival.

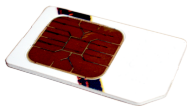
- Cooper was a Motorola engineer.
- In 1973 he called Joel Engel.
- Engel led AT&T's cellular team.
- AT&T was betting on car phones.
- Motorola bet on a handheld.
- Service reached users in the 1980s.

1G moved the telephone away from the wall.

The SIM card is the part that knows you



Nokia 3310



GSM SIM card

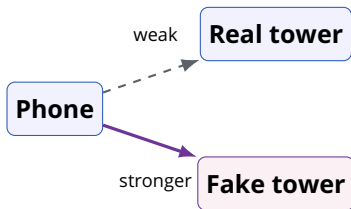
2G went digital.

- Digital voice and SMS
- Identity moved onto the SIM
- Mass-market roaming

SMS = Short Message Service SIM = Subscriber Identity Module

Q: Which one is “you” to the network: the phone, the SIM, or the number?

A: The SIM. It stores the subscriber identity and the secret key used to authenticate you. The number is only a label the network maps onto that identity.



Three steps.

- The phone attaches to the strongest signal.
- The tower asks who it is.
- The phone answers — normally with a temporary ID.

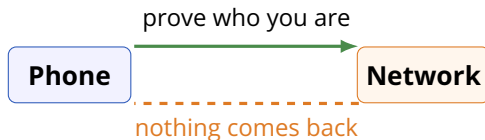
Unless the tower insists on the permanent one.

IMSI = International Mobile Subscriber Identity, the permanent one

Equipment built to insist has a name: an IMSI catcher.

This describes a capability and a class of equipment. It is not a claim about anyone in this room.

2G checked your ID. You never got to check theirs.



Q: The SIM proves the phone is real. What proved the tower was real?

A: In 2G, nothing — and the network also chose whether to encrypt at all. From 3G onward it must prove itself too, but the phone still has to pick a tower before it can check anything.

We trace what each generation did about this on Wednesday.

3G brought video calls before eye contact



Nokia 6680 commercial (YouTube)

Data became central.

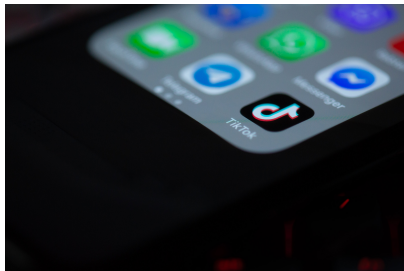
- Web, email, multimedia
- UMTS/WCDMA radio access
- Packet data grew beside voice

UMTS = Universal Mobile Telecommunications System

WCDMA = Wideband Code Division Multiple Access

3G made the phone a data device.
The core still ran voice and data as two separate worlds.

4G made buffering feel like a personal insult



TikTok app on phone

Broadband became the default.

- Streaming and short video
- Maps, ride sharing, live apps
- Voice moved to IP with VoLTE

VoLTE = Voice over LTE IP = Internet Protocol

4G did not invent video. It made video normal.

You press Call. What happens next?



UMTS video call

Your phone has to find a tower.
The network has to recognize you.
Then it has to find the other end.

Q: Which part of the network handles each of those three jobs?

A: The RAN finds the tower and holds the radio link. The core recognizes you and decides what you may do. The core also reaches the other end, through the Internet or the telephone network.

Four worlds are hiding behind one tap



RAN

Turns radio waves
into managed links.

Core

Remembers who you are,
where you are,
and what you may do.

UE = User Equipment (your phone) RAN = Radio Access Network

PSTN = Public Switched Telephone Network, the traditional telephone network

3GPP = 3rd Generation Partnership Project, the body that writes these standards

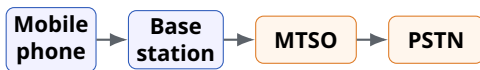
Conceptual model based on 3GPP system architecture

The network still has the same six jobs

- 1 Find a cell
- 2 Request access
- 3 Prove identity
- 4 Get radio resources
- 5 Route voice or data
- 6 Move without dropping

The node names change.
The jobs are surprisingly persistent.

1G connected radio voice to the telephone network



A circuit stays reserved
while the call is active.

MTSO = Mobile Telephone Switching Office

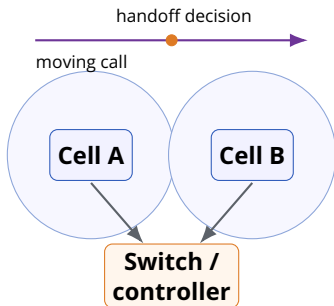
PSTN = Public Switched Telephone Network

The path was a circuit.

- Analog radio in the cell
- MTSO linked the towers
- PSTN carried the call

A mobile call was still a telephone call.

Mobility means changing the network while using it



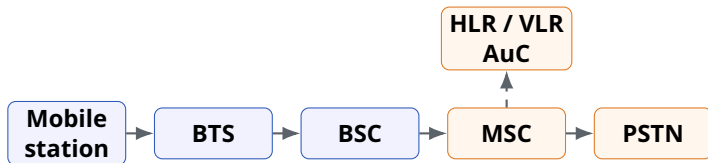
A moving car raises three questions:

- Who measures the signal?
- Who picks the next cell?
- Who keeps the call alive?

Q: What goes wrong if the handoff is late? If it is too early?

A: Late, the call drops while the old cell fades. Too early, the phone ping-pongs between cells and spends signaling on a move it did not need.

2G split the base station into radio and control



BTS Base Transceiver Station

BSC Base Station Controller

MSC Mobile Switching Center

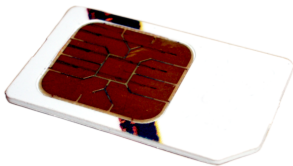
HLR / VLR Home / Visitor Location Register

AuC Authentication Center

Why split it? One BSC steers many BTSs. Radio hardware has to sit at every tower. Deciding to hand a call from one cell to the next has to see several towers at once. Those two jobs stopped fitting in one box.

Digital identities made roaming scalable.

The SIM became a network passport



GSM SIM card

Home network

knows your subscription, number, and credentials.

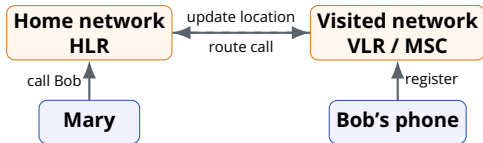
Visited network

knows where to reach you right now.

Q: If you replace the phone but keep the SIM, what changed in the network?

A: Almost nothing. The network still authenticates the same subscriber record, so your number, your subscription, and your roaming state survive. Only the device identity and its radio capabilities changed.

A call can find you even when you leave home



Bob visits another city.

- 1 Bob registers there.
- 2 The HLR is updated.
- 3 Mary's call is routed.

Mobility also needs a database.

GPRS was an Internet sidecar

GPRS = General Packet Radio Service, packet data added onto GSM.

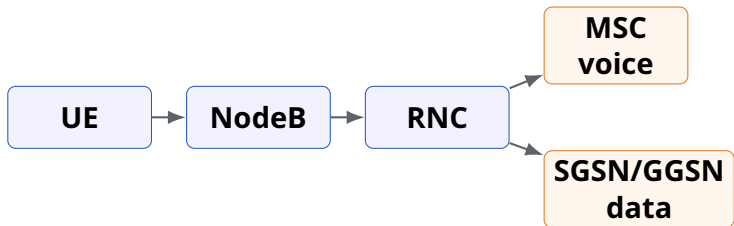


SGSN = Serving GPRS Support Node GGSN = Gateway GPRS Support Node

- Voice still used the MSC path.
- Packet data used SGSN and GGSN.
- The phone gained a standing route to the Internet.

Packet data arrived next to voice, not instead of it.

3G changed the radio faster than the architecture



NodeB = 3G radio node • RNC = Radio Network Controller

- UMTS/WCDMA raised mobile-data capability.
- NodeB and RNC echoed the BTS/BSC split.
- Circuit voice and packet data lived side by side.

[3GPP 3G overview](#)

Voice

Circuit switched

MSC and the telephone world

Data

Packet switched

SGSN/GGSN and the Internet

Running both worlds at once worked. It was not simple.

Can you match the tower name to the generation?

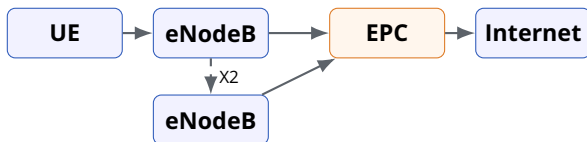
- | | |
|----------|-----------|
| ① BTS | • 2G GSM |
| ② NodeB | • 3G UMTS |
| ③ eNodeB | • 4G LTE |
| ④ gNodeB | • 5G NR |

GSM = Global System for Mobile Communications NR = New Radio
The prefixes are generational: e = evolved (LTE), g = next generation (5G).

Q: Which generation removed the separate radio controller?

A: 4G. The eNodeB absorbed the scheduling and handover control that the RNC used to own, so LTE has no separate controller box in the main path.

4G flattened the RAN

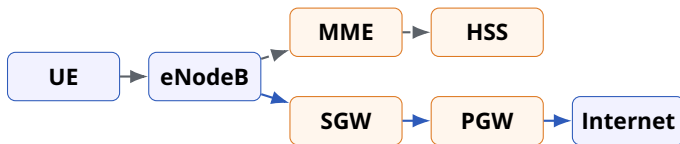


No separate RNC sits in the main LTE path. EPC = Evolved Packet Core

- The eNodeB absorbed the radio control that lived in the RNC.
- Neighboring eNodeBs coordinate directly over X2.
- The core became the Evolved Packet Core.

Fewer boxes in the path.
More logic inside the base station.

The EPC turned the core into packet plumbing



dashed = control signaling • solid blue = user data

MME Mobility Management Entity

HSS Home Subscriber Server

SGW Serving Gateway

PGW Packet Data Network Gateway

Learn the jobs, not the acronym list.

Wait — where did SGSN and GGSN go?

What it did

SGSN tracked where you were and kept your data session alive. GGSN was the door out to the Internet.

Each box did both jobs at once: it made decisions **and** it carried your traffic.

The clean idea arrived in one release. The clean **implementation** took nine more years.

What broke

Traffic grew far faster than decisions did.

Ten times the video does not mean ten times the number of attach requests. But because one box did both, carrying more data meant buying more decision-making you did not need.

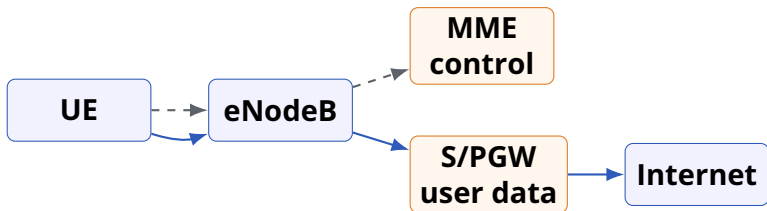
What replaced it

4G **started** the cut. MME decides and never touches your packets.

The gateways moved packets — but kept deciding too. Finishing the split took until 2017.

The box did not fail. It was doing two jobs with one growth rate.

Control asks; the user plane moves packets



Control plane

attach, authenticate,
move, establish a bearer

User plane

carry the voice, video,
game, or web packets

Separating the two planes
lets each one scale on its own.

And the MSC? It had been there since 2G.

What it did

The MSC was a telephone switch that happened to serve mobiles.

It set up a circuit for each call and held that circuit open until somebody hung up.

What broke

LTE's own bearers carried IP and nothing else.

A circuit switch has no place in a packet network. Keeping one meant running a second network beside the first, with its own equipment, staff, and bill.

What replaced it

At first, the MSC itself.

An LTE phone that wanted to call **dropped back** to 2G or 3G to do it. Texts still reached a legacy switch.

It was not replaced. It was demoted.

CSFB = Circuit-Switched Fallback, the standardized way an LTE phone borrowed an older network to make a call

So who finally carries your voice call on an all-IP network?

In 4G, a phone call had to become an Internet app

Early LTE

Data on LTE

Voice falls back to 2G/3G

VoLTE = Voice over LTE IMS = IP Multimedia Subsystem, the part of the core that sets up and controls calls

VoLTE

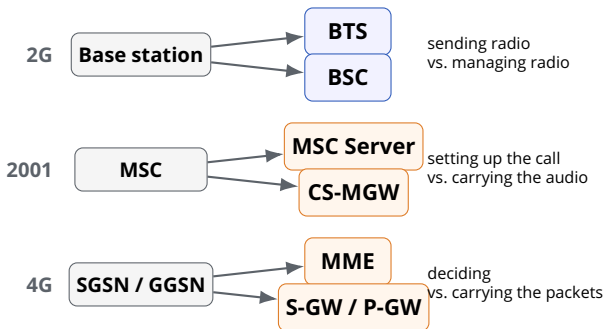
Voice as a managed IP service

IMS provides call control

Q: Why was “all-IP” a bigger change than making the radio faster?

A: Speed is one number. All-IP changed the structure: one packet transport carries everything, and voice becomes an application on top of it instead of its own circuit-switched network.

This is not a coincidence anymore



One box becomes two along the seam between deciding and carrying.

You will meet this seam twice more on Wednesday — in the 5G core, and in the base station itself.

Monday, in five sentences

- 1 The icon on your phone names a radio, and somebody pays about \$50 a month to put it there.
- 2 Every generation made one new habit feel normal: voice, text, web, video.
- 3 The network has always had the same jobs — find you, admit you, move your bits, bill you.
- 4 A box splits in two when its halves need to change at different speeds.
- 5 Your phone attaches to the loudest tower, and in 2G nobody checked whether that tower was honest.

Sentence four is the one that will keep paying off.

On one card, in your own words:

- 1 Name one box that disappeared between two generations, and say what pressure removed it.
- 2 You are on a 5G icon and the page still loads slowly. Give one reason that has nothing to do with the radio.

Wednesday: the core becomes software, the base station comes apart, and we find out whether anyone can tell a fake tower from a real one.

Part 2

How the network became software

Wednesday, 9/2

5G core · Security · O-RAN · Who pays for it

Two things from Monday you need today

The rule

One box becomes two when its halves need to change at different speeds.

You saw it twice: the 2G base station split into radio and control, and the GPRS core split into decisions and packets.

Today it happens twice more.

RAN = Radio Access Network Core = everything behind the towers

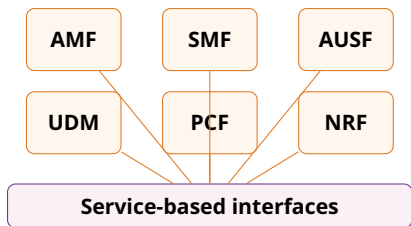
The loose end

Your phone attaches to the strongest signal it can hear.

In 2G it proved itself to the network, and the network proved nothing back.

Today we find out what four generations did about that.

5G made the core look more like a cloud platform



AMF Access + Mobility
SMF Session Management
UPF User Plane Function
AUSF/UDM identity
PCF policy
NRF service registry



The SMF steers the UPF over the N4 interface.

[Nokia 5G Core Service Based Architecture](#)

Core functions now discover and call each other as services.

Third time. Where did the MME go?

What it did

In 4G the MME was the one box that made control decisions: who may attach, where you are, which gateway serves you.

One box, every kind of decision.

What broke

Those decisions do not resemble each other.

Checking who you are happens once. Tracking where you are happens constantly. Setting up a session happens per app.

Same box, three completely different loads.

What replaced it

5G cut the control plane itself.

AMF handles access and mobility. SMF handles sessions. Each is a separate service that scales on its own.

AMF = Access and Mobility Management Function SMF = Session Management Function

Same rule, one level deeper: now the control plane splits.

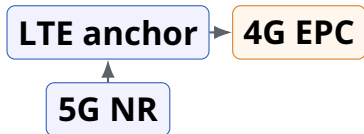
A 5G radio does not guarantee a 5G core

NSA = Non-Standalone

SA = Standalone

QoS = Quality of Service

NSA



The phone shows 5G.
Control still runs through
LTE and the 4G core.

SA

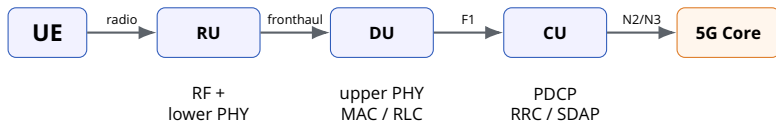


A 5G radio on a 5G core.
Slicing, edge, and new QoS
become possible.

The icon reports the radio.
The features come from the core.

AT&T: "Not all 5G is created equal," 2025

The 5G base station can be split across locations



One gNodeB can be a system spread across a tower, a cabinet, and a data center.

RU / DU / CU = Radio / Distributed / Central Unit

PHY = physical layer MAC = Medium Access Control RLC = Radio Link Control

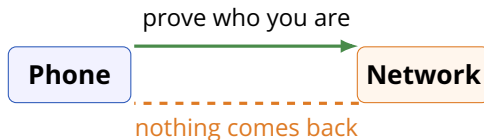
PDCP = Packet Data Convergence Protocol RRC = Radio Resource Control

SDAP = Service Data Adaptation Protocol

Q: Who owns each box, and who owns the interfaces between them?

A: In a classic RAN, one vendor owns all of it. Splitting the base station only helps if the seams between the pieces are specified well enough for a second vendor to plug in.

Monday's loose end



In 2G the phone proved itself to the network. The network proved nothing back, and the phone had already picked the loudest tower before anyone asked.

Every generation since has been trying to close that gap.
Did they?

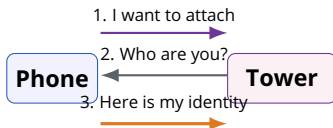
The same problem, four times

	The weakness	The answer
2G	Network never authenticates itself	—
3G	Identity still sent in the clear at first contact	Mutual authentication added
4G	Same first-contact exposure	Stronger keys, same opening move
5G	Phone must still pick a tower first	Identity encrypted before it is sent

Each generation closed a door. None of them closed the doorway.

Simplified for teaching. Each row is a family of changes across many releases, not a single switch that was flipped.

Why the first message is the hard one



Step 3 happens before the phone has verified anything about the tower.

The bootstrap problem.

To check a tower's credentials, the phone needs a channel.

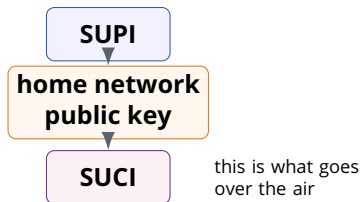
To get a channel, it has to talk to the tower.

Something has to go first.

IMSI = International Mobile Subscriber Identity

SUPI = Subscription Permanent Identifier

5G stopped sending the identity in the clear



SUCI = Subscription Concealed Identifier

Q: So is the problem solved?

A: Partly. It closes the easy collection of permanent identities — but only where the operator has actually turned it on, and only if the phone stays on 5G.

Encrypt it first.

The permanent identity never leaves the phone unprotected.

What the tower receives is a concealed version that only the home network can open.

Two ways it still leaks

Configuration

Concealment is a feature operators enable.

A network that has not turned it on gets none of the benefit, and the phone cannot tell from the outside.

Downgrade

Phones still support older generations so they keep working where 5G does not reach.

Push a phone down to a weaker generation and you are back to that generation's rules.

Backward compatibility is a feature and an attack surface at once.

Downgrade behavior against 5G devices is demonstrated in the research literature. Treat it as a known capability, not as a claim about any particular network.

What would it take to notice?

- A real tower and a fake one look similar from one measurement.
- The signal that matters is a **pattern**: odd parameters, a cell that appears from nowhere, a push to an older generation.
- Seeing a pattern means watching many cells at once, over time.
- And the pattern changes faster than the standards do.

Where in the network could that detector even live?

Hold that question. It is the one the next section was built to answer.

Why this belongs in an architecture lecture

Every fix in this section was a change to **who talks to whom, in what order, and who is allowed to decide.**

That is not a security topic bolted onto an architecture topic.
It is the same topic.

Architecture decides what an attacker is allowed to try.

Can base-station security become an app?

New defense

Vendor ticket

New image

Maintenance
window

You invent a defense against a new attack on the base station.

Q: Who installs it, who approves it, and what happens at 2:00 a.m.?

A: In a classic RAN the vendor answers all three. Your idea becomes a support ticket, then a firmware image, then a scheduled maintenance window — and nothing moves fast at 2:00 a.m.

O-RAN's bet:
some network behavior can ship as a tested application.

A traditional RAN often arrived as a sealed appliance



- One vendor shipped the box.
- One vendor owned the debugging.
- Upgrades followed its train.

Easy to blame one box. Hard to change one piece.

Telenor radio equipment rack

Four RAN models, four different questions



Virtualization

What runs where?

D-RAN = Distributed RAN C-RAN = Centralized (or Cloud) RAN
vRAN = Virtualized RAN O-RAN = Open RAN

Disaggregation

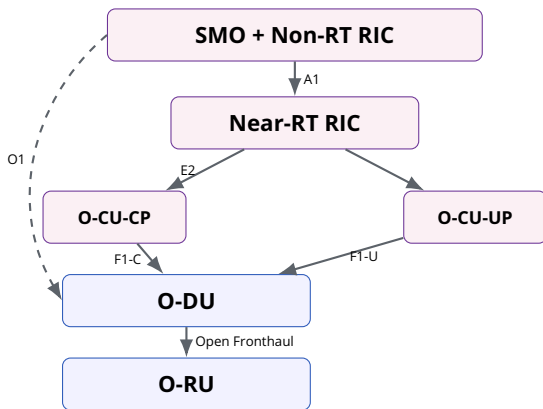
Which functions split?

Open interfaces

Who can connect?

Cloud software alone does not create an open,
multi-vendor RAN.

O-RAN opens the seams, then adds control loops



RAN chain

O-CU-CP / O-CU-UP
O-DU / O-RU

Control

SMO + Non-RT RIC
Near-RT RIC

Open seams

A1, E2, O1, Open
Fronthaul

SMO = Service Management
and Orchestration
RIC = RAN Intelligent
Controller

Non-RT / Near-RT =
non- and near-real-time
CP / UP = control / user plane

O-RAN ALLIANCE architecture, redrawn from 2023 white paper

Open RAN is not the same as open-source RAN

Open RAN

Published interfaces
Disaggregated functions
Multi-vendor is possible

Open source

Source code is available
License permits reuse
Interface may stay closed

Q: If two boxes claim the same interface, will they work when plugged together?

A: Not automatically. A specification leaves options, versions, and timing choices open. Conformance and interoperability testing — plugfests, lab integration — is what finishes the job.

[O-RAN ALLIANCE architecture and resources](#)

RIC makes optimization an application



[Play: a real RIC demo \(0:18–1:12\)](#)

[ITU, O-RAN / RIC demonstration](#)

Non-RT RIC + rApp
policy and models, seconds and up

↓ A1

Near-RT RIC + xApp
faster decisions: roughly 10 ms–1 s

↓ E2

O-CU / O-DU / O-RU

Example: traffic steering

The rApp sets the policy.

The xApp reacts to load.

The RAN executes.

rApp / xApp = apps on the
Non-RT / Near-RT RIC

Opening the RAN creates both options and work

Options

- More vendor choice
- Faster experiments
- Programmable control

Work

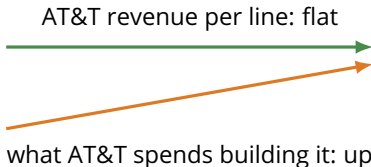
- Fronthaul and timing
- Integration testing
- Bigger security surface

Q: Would you rather debug one black box or five open interfaces?

A: Both are work, in different places. One box hides the failure and hands the whole fix to one vendor. Five interfaces make the failure visible, and make the integration yours.

[O-RAN ALLIANCE overview \(2026\)](#)

Monday's open question



Two lines going different directions.

How long can that last, and what happens at the end of it?

Roughly where a fifth of your bill goes

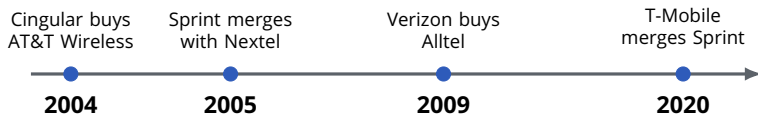
$$\frac{\$17 \text{ billion}}{126 \text{ million}} \text{ a year of network construction}$$
$$\div$$
$$\frac{126 \text{ million}}{\text{postpaid connections}}$$

$$\approx \$11 \text{ per connection per month}$$

Against a bill of roughly \$50, that is about a fifth of what you pay going straight back into towers, spectrum, fiber, and power.

Deliberately crude: consolidated capex covers broadband and business too, and “connections” counts tablets and watches, not just phones. Verizon FY2025 actual capex and Q1 2025 connections.

Four national carriers became three



Years shown are when each deal **closed**. Brands lingered for years after, which is why you still hear some of these names.

This is a market that consolidates, not one that fragments.

So are they in trouble, or are they fine?

Fine

- Three players, high barriers
- Spectrum licenses are scarce
- Everyone needs a phone
- Steady, predictable revenue

In trouble

- Revenue per line is not growing
- Every generation costs billions
- Customers switch on price
- Debt from spectrum and mergers

Q: Which of these is a network architecture problem?

A: Most of them. Cheaper coverage, shared infrastructure, and software you can upgrade without replacing hardware are all architecture answers to a financial question. That is a large part of why O-RAN exists.

The competitor already in your pocket

Wi-Fi.

Every hour you spend on home or campus Wi-Fi is an hour the carrier does not have to build capacity for.

That is help — and it is also a ceiling on what they can charge for data you were never going to buy from them.

When did you last care which one you were on?

The answer is usually: when something broke.

A competitor you do not notice is still a competitor.

Starlink direct-to-cell.

- Ordinary phones, no special hardware
- Started as a carrier partnership
- Then began looking like a rival

June 2026.

On the news that SpaceX might go direct to consumers, carrier shares fell sharply in a single session.

T-Mobile's CEO called the threat exaggerated.

As of late June 2026. This story is two months old and still moving — check what has happened since.

Who is right?

What a satellite can and cannot take

Satellites win

Where there are no towers and never will be: oceans, deserts, mountains, disasters.

Coverage is their whole argument.

Towers win

Where the people are. A stadium needs capacity in one small place, and capacity comes from cells close together.

One satellite covers a huge area — and everyone in it shares.

Coverage and capacity are different products. So far they are losing to each other on different ground.

The question to leave with

Every architecture change in these two lectures was somebody trying to serve more people for less money per person.

Splitting boxes. Moving voice onto IP. Turning the core into software.

Opening the RAN so you can buy the pieces separately.

If that pressure never stops, what does the network look like next?

Five generations in one low-resolution map

	Radio	RAN control	Core	Mode	Example
1G	Base station	MTSO	Analog core	Circuit	Voice in a car
2G	BTS	BSC	MSC + HLR/VLR	Circuit; GPRS packet	SMS, roaming
3G	NodeB	RNC	CS + PS cores	Both	Mobile web / video
4G	eNodeB	In the eNodeB	EPC, all-IP	Packet	Maps, apps, streaming
5G	gNodeB	CU (RRC), split CU/DU/RU	Service-based 5GC	Packet + QoS flows	Flexible services + edge

CS = circuit-switched PS = packet-switched QoS = Quality of Service
EPC = Evolved Packet Core 5GC = 5G Core

The radio name changes.
The deeper story is where control lives.

Follow the same action one last time



The packet still needs **radio access**,
identity + policy + routing,
and now sometimes **programmable control**.

Each generation relocates functions.
O-RAN changes who may control them.

Exit ticket: draw the boundaries



- 1 Mark the **RAN** and **Core** boundary.
- 2 Add one **O-RAN** interface or controller.
- 3 Answer in one sentence:
What changes next—radio, core, or control?

Appendix: the acronym decoder ring

Era	Node	Meaning / job
1G	MTSO	Mobile Telephone Switching Office; switching and mobility control
2G	BTS / BSC	Base Transceiver Station / Base Station Controller
2G/3G	MSC, HLR, VLR, AuC	Voice switch; home/visitor state; authentication
GPRS	SGSN / GGSN	Packet mobility + gateway to external packet networks
3G	NodeB / RNC	Radio node / Radio Network Controller
4G	eNodeB	LTE radio node; absorbs most RNC control
4G	MME, HSS, S-GW, P-GW	Mobility; subscriber data; serving and packet gateways
5G	gNodeB	5G radio node, often split into CU / DU / RU
5GC	AMF, SMF, UPF	Access/mobility; session control; packet forwarding
5GC	AUSF, UDM, PCF, NRF	Authentication; subscriber data; policy; service registry
O-RAN	SMO / RIC	Management/orchestration and intelligent control

Generic terms:

3GPP = 3rd Generation Partnership Project IP = Internet Protocol

QoS = Quality of Service PDU = Protocol Data Unit

CS / PS = circuit- / packet-switched CP / UP = control / user plane

Appendix: interface map

Uu UE ↔ radio: air

S1 / NG RAN ↔ core

X2 / Xn neighbor handover

F1 O-CU ↔ O-DU

Open FH O-DU ↔ O-RU

E2 Near-RT RIC ↔ RAN

A1 Non-RT RIC ↔ Near-RT RIC

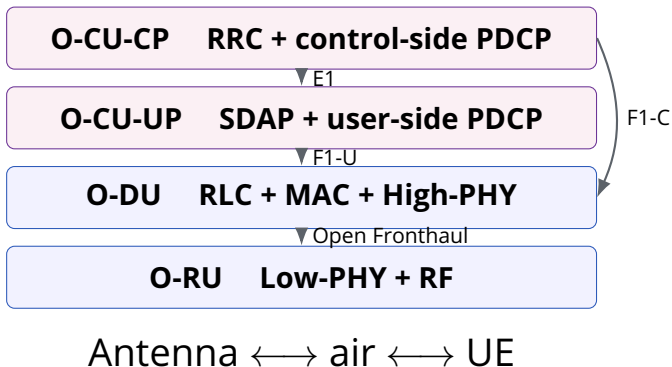
O1 SMO ↔ managed functions

O2 SMO ↔ cloud platform

Interfaces expose coordination;
they do not remove it.

[O-RAN ALLIANCE architecture and resources](#)

Appendix: where the protocol stack is split



A split is a placement decision.
Latency, bandwidth, compute, and ownership all move with it.

Appendix: handover changed owners, not purpose

Era	Main controller	What crosses the boundary?	What must survive?
1G	MTSO	Analog voice circuit	The conversation
2G	BSC / MSC	Radio channel + circuit state	Call and subscriber context
3G	RNC / core	Radio bearer; CS or PS state	Voice or packet session
4G	eNodeB + EPC	IP bearer and mobility context	All-IP session
5G	gNodeB + 5GC	PDU session, QoS flows, policy	Service continuity
O-RAN	Same 3GPP functions	RIC may influence parameters	Safe control under open APIs

Q: Which is harder to move—a radio channel, a packet tunnel, or its policy?

A: The channel has the hardest deadline; the policy is the hardest to keep consistent across cells, cores, and vendors.

Sources and further reading

- **3GPP TS 23.501**

Authoritative 5G system architecture.

- **Nokia: 5G Core service-based architecture**

Readable service-based-architecture role map.

- **O-RAN Architecture and Resources**

Interfaces, RICs, and current specifications.

- **O-RAN ALLIANCE overview (2026)**

Current official overview.

- **AT&T 5G Standalone announcement (2025)**

Why a 5G icon can appear before a 5G core.

- **Course references**

Local materials in `reference/` were used for background only, not reproduced as slide images.

Image licenses and page-level provenance: `figure/SOURCES.md`